# A factorised physics prior for Boussinesq-trained surrogates: the compressible correction in buoyancy-driven enclosure convection is linear to leading order in the relative temperature contrast

Abstract

Boussinesq-trained surrogates for thermal convection fail across temperature contrast because they learn one correction field and have no signal with which to rescale it. For steady two-dimensional buoyancy-driven enclosure convection we show that the compressible correction factorises to leading order as $D = \varepsilon P(X) + O(\varepsilon^2)$, where $\varepsilon = \Delta T/T_0$ and the network learns only the spatial shape $P$. Over 627 paired solutions spanning $\varepsilon = 0.02$–$1.9$, the relative correction is $R = A\varepsilon(1 + \kappa\varepsilon)$ with $\kappa = 0.42$–$0.83$; a free power law returns $m = 1.023 \pm 0.018$ and $0.995 \pm 0.019$ for $(u_x, u_y)$, pressure gives $1.009 \pm 0.015$ in the fixed gauge, and $\theta$ has no identifiable exponent. $Fr^2 = \varepsilon$ identically, and $Ma$ enters only at $O(Ma^2)$ in the retained low-Mach equations. The design rule makes a single-$\varepsilon$ public data set reusable: the channel-averaged extrapolation error falls from 0.131 to 0.086 (1.5×) within the boundary-condition family, though channel-averaged across families it does not beat doing nothing. For heat transfer the field norm is misleading: at $\varepsilon = 1.2$ the integral Nusselt number changes by 0.67% while the local wall flux changes by 4.25%, and a prior trained on the same data leaves a 2.56% integrated wall-$Nu$ error, worse than the 0.67% of doing nothing. A wall-flux penalty removes this (0.11% at $\varepsilon = 1.2$) and transfers to unseen contrasts, giving 0.17/0.20/1.21% at $\varepsilon = 0.4/0.8/1.6$ against 0.45/1.08/4.08% for doing nothing; at $\varepsilon = 1.9$ the gain is not resolved. On a GPU one inference takes 6.3 ms; replacing the compressible member saves 2.9× per pair and 13× for a 20-flow × 5-contrast scan. A 24-sample 3-D check at $Ra = 10^4$ and $3 \times 10^4$ ($res = 16$) returns exponents for $u_x$, $u_y$ and $p$ within 0.15 of unity.

Keywords: natural convection, Boussinesq approximation, low-Mach compressible flow, physics-informed surrogate, neural operator, parametric dependence

Nomenclature

| Roman | | Greek | |
|---|---|---|---|
| $A$ | prefactor of $R = A\varepsilon^m$ | $\beta$ | thermal expansion coefficient |
| $D$ | correction $Y - X$, the surrogate target | $\gamma$ | ratio of specific heats |
| $e$ | internal energy | $\Delta T$ | wall-to-wall temperature contrast |
| $E$ | total energy | | |
| $Fr$ | Froude number, $Fr^2 = \Delta T/T_0$ | $\varepsilon$ | relative contrast, $\Delta T/T_0$ |
| $\boldsymbol{g}$ | gravitational acceleration | | |
| $L$ | side length of the cavity | $\theta$ | normalised temperature |
| $m$ | scaling exponent | $\kappa$ | curvature, $R = A\varepsilon(1 + \kappa\varepsilon)$ |
| $Ma$ | Mach number | $\rho$ | density |
| $n$ | number of samples in a data set or split | $\boldsymbol{\sigma}$ | viscous stress tensor |
| | | $\chi$ | base-flow advection strength |
| $Nu$ | Nusselt number | | |
| $p$ | pressure | | |
| $P$, $Q$ | first- and second-order correction fields | | |
| $\boldsymbol{P}_{\text{work}}$ | viscous work term | | |
| $Pr$ | Prandtl number | | |
| $q$ | Rayleigh exponent, $R \propto Ra^q$ | | |
| $R$ | relative correction, $\|Y - X\|/\|Y\|$ | | |
| $Ra$ | Rayleigh number | | |
| $Ra_{\text{eff}}$ | effective Rayleigh number, $Ra\,Fr^2/\varepsilon$ | | |
| $Re$ | Reynolds number, $(Ra/Pr)^{1/2}$ | | |
| $T$ | temperature, $T_0$ its reference value | | |
| $\boldsymbol{u}$ | velocity, $(u_x, u_y)$ | | |
| $\hat{u}$ | peak Boussinesq speed | | |
| $X$, $Y$ | Boussinesq and compressible solutions | $\mathcal{L}$ | linearised low-Mach operator at $O(\varepsilon)$ |

## 1. Introduction

Steady natural convection is a workhorse of thermal engineering and it is expensive. A converged pair of two-dimensional solves on a $64^2$ mesh costs 34 s of wall clock on an eight-way desktop (Sec. 5.12); the three-dimensional problems that engineering actually needs cost hours to days. That is the standard motivation for surrogate modelling, and the standard response is to train a network that maps a cheap model's output to an expensive model's output [1, 2].

For convection the two models are not two arbitrary fidelities: they are the Boussinesq (Oberbeck–Boussinesq) equations and the compressible low-Mach equations. Boussinesq is exact only in the limit of vanishing relative temperature contrast; the corrections are $O(\beta\Delta T)$ and have been characterised asymptotically for decades [3, 4]. Recent work makes the pair directly learnable. Menglik et al. [5] publish paired Boussinesq→compressible steady solutions in two dimensions (4000 training pairs plus held-out splits, CC BY 4.0) and train a Fourier neural operator [6] to predict the compressible field from the Boussinesq field. That data set—which we use as an independent validation set in Secs. 5.7–5.9—contains exactly one temperature contrast: $\Delta T = 1.2$ at $T_0 = 1$.

The single-contrast design is the practical crux. A surrogate fitted on such a data set has learned the correction at one operating point; at any

other contrast the Boussinesq input $X$ is unchanged, so the network has no signal with which to rescale what it memorised, and the only remedy on offer is another paired data set—paying again the cost the surrogate was meant to remove. Sec. 5.3 shows that this is not a capacity problem: the missing variable is $\varepsilon$, and we measure the resulting ceiling.

This paper makes the barrier explicit and then removes it.

What governs the gap. For the non-dimensionalisation used here the Froude number is not free: $Fr^2 = \beta\Delta T = \Delta T/T_0$ identically, and the Mach number enters the equations only at $O(Ma^2)$ within the retained low-Mach equations (Sec. 2.2; a factor-24 change in $Ma$ leaves the measured correction unchanged at solver precision, which checks the implementation against the retained equations rather than establishing physical $Ma$ independence). Within the retained equations and this scaling there is therefore one free parameter rather than two, and it has a name: $\varepsilon = \Delta T/T_0$.

What the gap looks like. Expanding the compressible member in $\varepsilon$ gives an $O(1)$ system that is term-by-term the Boussinesq system, plus an $O(\varepsilon)$ source produced by the density-modulated inertia and by the non-solenoidal velocity field $\nabla \cdot \boldsymbol{u}_1 = \nabla \cdot (\theta_0 \boldsymbol{u}_0)$. Because the linearised operator is $\varepsilon$-independent, the correction factorises, $Y - X = \varepsilon P(X) + \varepsilon^2 Q(X) + \ldots$, with a shape $P$ that does not rotate as $\varepsilon$ moves (Sec. 3). Measured over two decades of $\varepsilon$ (627 samples, Sec. 5.1) the relative correction obeys $R = A\varepsilon^m$ with $m = 1.023 \pm 0.018$ for $u_x$ and $0.995 \pm 0.019$ for $u_y$. The pressure channel gives $m = 1.01$ in the fixed gauge of Sec. 5.2, a numerical observation rather than a gauge-invariant exponent. $\theta$ is a weak, flow-dependent channel: its free power-law fit is not identifiable ($R^2 = 0.22$; slope 1.02 over the full window, 1.67 on the 104-configuration subset), so we make no exponent claim for it; its amplitude is set by the flow, with the $\chi$ parameterisation at $R^2 = 0.78$ and the $\hat{u}$ form at 0.90 (Sec. 5.1). The exponent is window-averaged: the same 627 points give 1.23–1.32 when fitted from $\varepsilon = 0.8$ upwards, which is the $\varepsilon^2$ curvature. The intercept vanishes, and the prefactor's weak Rayleigh-number trend reproduces in a second geometry and in a third Prandtl number.

What it buys. Because the $\varepsilon$ dependence can be written outside the network, a surrogate can predict $P(X)$ and multiply by $\varepsilon$ afterwards. Supplying $\varepsilon$ as an input as well gives the largest gain relative to the strongest conditional baseline: 1.07× in $u_x$ for the linear form and 1.15× for the quadratic form (channel-averaged 1.09× and 1.21×; the $\theta$ and $p$ channels are individually significant, $p$ in the fixed gauge, while the $u_x$ and $u_y$ intervals include zero; Sec. 5.5)—while removing it from the input raises the extrapolation error from 0.0937 to 0.1214 in $u_x$ and from 0.1126 to 0.1682 in $p$ (Table 5), but that removal is what lets a network trained on a public data set that exists at a single $\varepsilon$ be applied, without retraining, to our $\varepsilon \in [0.2, 1.9]$, where it

reaches relL2 = 0.093 ($u_x$) against 0.144 for the plain model at the same budget (Sec. 5.9). The prior is also the cheaper model when data is scarce (Sec. 5.6: 1.4× better than an $\varepsilon$-augmented baseline at $n = 50$, crossover near $n \approx 300$), the curvature it learns is positive in all four channels ($\kappa_{\text{net}} = 0.22$–$0.42$, smaller than the 0.42–0.83 of the direct quadratic fit but of the same order), and the same prior transfers to a Fourier neural operator with the same qualitative pattern ($u_x$ unchanged, $p$ 13% worse than with an explicit $\varepsilon$ input).

### 1.1. Related work

Reduced-order models replace the discretised equations by a few modes [7]; graph convolutional networks exploit the mesh and have been applied to natural convection in annuli [8] and to geometry-adaptive steady convection [9]. Operator learning removes the mesh constraint: DeepONet approximates the solution operator from branch and trunk networks [10], physics-informed DeepONets add the residual to the loss [11], and Fourier neural operators (FNOs) learn a global convolution in Fourier space [6], with applications from sparse-field reconstruction [12] to heat-pipe startup [13] and uncertainty quantification in cooling-channel optimisation [14]. Physics-informed neural networks [15, 16, 17] instead solve one instance by minimising a PDE residual. Reviews are given by De Ryck and Mishra [18], Meng et al. [2], Luo et al. [19].

A physics prior can enter as a residual penalty, as a conservation law built into the architecture, or as an algebraic factorisation of the output. Residual-based corrector operators solve a linear variational problem for the residual of a trained operator and add the solution to its prediction; the correction is deterministic, costs one linear solve per query, and needs no retraining [20, 21]. Physics-informed extrapolation adds sparse observations or residual terms [22], transfer operator learning addresses distribution shift [23], and multi-fidelity surrogates combine cheap and expensive data [24, 25, 26]. These methods reduce the high-fidelity data requirement but still learn the parameter dependence from samples.

For buoyant convection the Oberbeck–Boussinesq approximation and its limits are classical [3, 4, 27]; non-Oberbeck–Boussinesq effects have been measured under strong contrast [28], in cryogenic helium [29] and in water [30], and at $Pr = 0.71$ they can reverse the large-scale circulation through cold-corner vortices [31]. Low-Mach formulations for enclosed convection have been compared with the Boussinesq limit [32, 33], and the rigorous derivation has been revisited [34]; that limit carries a non-local boundary condition for the temperature deviation, whereas the expansion here uses the retained low-Mach system with fixed wall temperature, where the deviation vanishes at every order on the walls. What is missing is a quantitative account of how the difference between the two solutions scales with the

contrast, and a surrogate design that uses it: these studies fix $\varepsilon$ and compare the two solutions, none extracts the $\varepsilon$ dependence as a reusable prior.

Table 1 places the resulting prior among the four families.

Contributions.

1. We show that, in the non-dimensionalisation used here and within the retained low-Mach equations, the Boussinesq–compressible gap is controlled by one parameter, $\varepsilon = \Delta T/T_0$, rather than by $\varepsilon$ and $Ma$ independently (Sec. 2.2).
2. We derive the correction's factorisation from an $\varepsilon$-expansion and verify all five of its predictions numerically, including the divergence relation (Sec. 3).
3. We calibrate the resulting law on 627 paired samples spanning two decades of $\varepsilon$ in the unit square at $Pr = 0.71$, test it in a second geometry (104 samples) and at $Pr = 4.38$ (104 samples), and validate the prefactor on an independently generated public data set—including the secondary Rayleigh-number trends (Secs. 5.1, 5.7, 5.10, 5.13).
4. We measure what the gap does to the wall heat transfer: at $\varepsilon = 1.2$ the integral Nusselt number moves by 0.67% (ratio of ensemble means; per-sample mean 0.54%, median 0.58%) on the public data, while the local wall flux is redistributed by 4.25%; on the five isothermal-sidewall configurations the integral change at $\varepsilon = 1.9$ is $-5.1\%$ at $Ra = 10^4$ and $-1.0\%$ at $Ra = 10^3$. The 15 random-wall configurations are not used for a relative $Nu$ statistic: eight of them have $|Nu_X| \leq 1$, where the ratio is dominated by its denominator (Sec. 5.8).
5. We turn the law into a surrogate design rule ($D = \varepsilon P(X)$), measure what it is worth (1.09× lower channel-averaged error than the strongest conditional baseline in extrapolation, with $\theta$ and $p$ individually significant ($p$ in the fixed gauge) and the $u_x/u_y$ intervals including zero, and indistinguishable within seed scatter in interpolation), and show it makes an existing single-$\varepsilon$ public data set reusable, with the end-to-end saving measured per pair (2.9×), for the 20-flow × 5-contrast scan (13×) and for a single flow swept over 100 contrasts ($2.3 \times 10^2$) (Secs. 5.4–5.9, 5.12).

All results are two-dimensional except the twenty-four-sample three-dimensional check of Sec. 5.14 (three wall types at two Rayleigh numbers). Section 6 states the measured cost of the three-dimensional problem on the same hardware and gives the concrete reasons we do not claim a 3-D sweep.

Table 1: Where the present prior sits among physics-informed operator methods. "Derived" is obtained in closed form from the equations; "learned" is supplied by the network. Rows: post-hoc correctors [20, 21], physics fine-tuning and multifidelity [22], transfer operator learning [23], dimension-informed operator [35].

| family | derived | learned | target data | per query | law |
|---|---|---|---|---|---|
| post-hoc correctors | linear residual | none | no | one linear solve | no |
| physics fine-tuning | residual / data | correction | yes | no | no |
| transfer operator | none | operator | yes | no | no |
| dimension-informed | equivariance | operator | no | no | no |
| this work | $\varepsilon$ expansion | shape $P$ | no | no | $m$, $\kappa$ |

## 2. Problem, discretisation and the break-symmetry parameter

### 2.1. Two members per sample

Every sample is a paired steady solution of one physical configuration with two equation sets on the same mesh ($res = 64$, $L = 1$, $Pr = 0.71$, $\gamma = 1.4$). The high-fidelity member $Y$ is the compressible low-Mach formulation in $\boldsymbol{w} = (\boldsymbol{u}, T, \rho, p)$:

$$\nabla \cdot (\rho \boldsymbol{u}) = 0, \tag{1}$$

$$\nabla \cdot (\rho\, \boldsymbol{u} \otimes \boldsymbol{u}) = -\nabla p + \frac{1}{Re}\nabla \cdot \boldsymbol{\sigma} - \frac{1}{Fr^2}\rho \boldsymbol{g}, \tag{2}$$

$$\nabla \cdot (\rho E \boldsymbol{u}) = \frac{\gamma}{Re\, Pr}\nabla^2 e + \nabla \cdot (\boldsymbol{P}_{\text{work}}\boldsymbol{u}) + O(Ma^2), \tag{3}$$

$$\rho e = \gamma Ma^2 p + 1, \qquad Re = (Ra/Pr)^{1/2}. \tag{4}$$

The low-fidelity member $X$ is the Boussinesq system in $(\boldsymbol{u}, T, p)$, written in the same free-fall scales:

$$\nabla \cdot \boldsymbol{u} = 0, \tag{5}$$

$$\boldsymbol{u} \cdot \nabla \boldsymbol{u} = -\nabla p + \left(\frac{Pr}{Ra}\right)^{1/2} \nabla^2 \boldsymbol{u} + \theta \hat{\boldsymbol{g}}, \tag{6}$$

$$\boldsymbol{u} \cdot \nabla T = (Ra\, Pr)^{-1/2}\nabla^2 T, \qquad \theta = \frac{T - T_0}{\Delta T}. \tag{7}$$

Both are solved with Taylor–Hood-type mixed elements, damped Newton iterations and a MUMPS factorisation; steady states are reached by an adaptive march with a convergence guard. Both members of a pair start from the same initial field and the same wall-temperature seed, so they follow the same branch. The temperature boundary data are random Fourier wall temperatures with per-wall mode dials; the classical differentially heated cavity (constant hot and cold side walls, adiabatic top and bottom) is the

one-mode member of the same family. Both are buoyancy-driven enclosure flows, and the results below are not restricted to the Rayleigh–Bénard layer.

The generator draws $Ra$ and $\varepsilon$ independently. At fixed $Ra$, changing $\varepsilon$ changes $Fr = \sqrt{\varepsilon}$, i.e. the buoyancy scale, while $Re = (Ra/Pr)^{1/2}$, $Pr$ and $\gamma$ stay fixed. The Boussinesq member therefore sees only $Ra$ and the thermal boundary data; the compressible member sees the same flow at a different $Fr$. This is a statement about the dimensionless parameterisation, not a claim that one fixed apparatus realises every pair: at fixed $Ra$, changing $\varepsilon$ requires the dimensional grouping $g\beta L^3/(\nu\alpha)$ to change in inverse proportion to $\Delta T$. A fixed apparatus follows the one-dimensional slice $Ra \propto \varepsilon$; the $(Ra, \varepsilon)$ rectangle contains that slice but is not restricted to it.

### 2.2. $Fr$ is not a parameter, $Ma$ is negligible, $\varepsilon$ is the axis

Table 2 lists the three candidates for a break-symmetry axis and what rules each one out.

Table 2: The three candidate break-symmetry parameters. Only $\varepsilon$ survives as a free axis in the present non-dimensionalisation. The $Ma$ row is a consistency check on the retained equations, which drop the $O(Ma^2)$ term.

| candidate | evidence | conclusion |
|---|---|---|
| $Fr$ | the generators state $Fr^2 = \beta\Delta T = \Delta T/T_0$ identically; the released 2-D code pinned $Fr = 1.095$, correct only at $\Delta T = 1.2$ | not free; pinned-$Fr$ sweeps of $T_0/\Delta T$ are an artefact |
| $Ma$ | the generator sets $Ma = 1.78\times10^{-3}(Ra/10^6)^{1/6}$, i.e. $3.8\times10^{-4}$ at $Ra = 10^2$ and $1.8\times10^{-3}$ at $Ra = 10^6$; fixing $Ra$, $T_0$, $\Delta T$ and changing $Ma$ by 24× leaves the measured correction unchanged to a relative $10^{-5}$ | $O(Ma^2)$ is dropped by the retained equations; the sampled range does not resolve it |
| $\varepsilon = \Delta T/T_0$ | same boundary-condition seed stream, $\Delta T$ swept | the clean, one-dimensional break-symmetry axis |

### 2.3. The low-fidelity solution is exactly $\varepsilon$-blind

$\theta$, $Pr$ and $Ra$ are the only quantities that appear in the Boussinesq weak form; neither $T_0$ nor $\Delta T$ appears alone. Hence, with the thermal data prescribed in $\theta$ and $Ra$ fixed, $X$ cannot depend on $\varepsilon$. This was verified to solver precision: across 20 configurations × 5 values of $\varepsilon$ (same $Ra$, bit-identical $\theta$ boundary stream), the maximum relative deviation of $X$ is $9.77\times10^{-8}$.

> Corollary. The amplitude of the correction is set by $\varepsilon$, which is absent from the surrogate's input. The spatial shape of the correction is a function of $X$ and is learnable; no function of $X$ alone, however expressive, can supply the factor $\varepsilon$ that scales it. The limit is informational, not architectural (Sec. 5.3).

3. Asymptotic theory: why the correction factorises as $\boldsymbol{\varepsilon P(X)}$

Write $T = T_0(1 + \varepsilon\theta)$ with $\theta = O(1)$. With $\rho = 1/T$ (the equation of state and initial condition used here) and $Fr^2 = \varepsilon$, the buoyancy source is

$$\frac{\rho \boldsymbol{g}}{\varepsilon} = \frac{\boldsymbol{g}}{\varepsilon} - \theta \boldsymbol{g} + \varepsilon\theta^2 \boldsymbol{g} + \ldots, \tag{8}$$

the constant $\boldsymbol{g}/\varepsilon$ is balanced by a hydrostatic pressure $p_{-1}/\varepsilon$ with $\nabla p_{-1} = -\boldsymbol{g}$, which drops out at the next order, so the $O(1)$ balances are

$$\nabla \cdot \boldsymbol{u}_0 = 0, \tag{9}$$

$$\boldsymbol{u}_0 \cdot \nabla \boldsymbol{u}_0 = -\nabla \tilde{p}_0 + Re^{-1} \nabla^2 \boldsymbol{u}_0 + \theta_0 \hat{\boldsymbol{g}}, \tag{10}$$

$$\boldsymbol{u}_0 \cdot \nabla \theta_0 = (Re\, Pr)^{-1} \nabla^2 \theta_0, \tag{11}$$

which is term-by-term the Boussinesq system solved as $X$ ($Re^{-1} = (Pr/Ra)^{1/2}$). The correction at $O(\varepsilon)$ is driven by the density modulation:

$$\nabla \cdot \boldsymbol{u}_1 = \nabla \cdot (\theta_0 \boldsymbol{u}_0), \tag{12}$$

$$\nabla \cdot (\boldsymbol{u}_1 \otimes \boldsymbol{u}_0 + \boldsymbol{u}_0 \otimes \boldsymbol{u}_1 - \theta_0 \boldsymbol{u}_0 \otimes \boldsymbol{u}_0) = -\nabla \tilde{p}_1 + Re^{-1} \nabla^2 \boldsymbol{u}_1 + \theta_1 \hat{\boldsymbol{g}} - \theta_0^2 \hat{\boldsymbol{g}}. \tag{13}$$

The energy equation contributes at the next order. Writing $T = T_0(1 + \varepsilon\theta_0 + \varepsilon^2\theta_1 + \ldots)$, the $O(\varepsilon)$ balance is the base energy equation for $\theta_0$ already contained in $\mathcal{L}$, and $\theta_1$ enters at $O(\varepsilon^2)$ in $T$; the weight $1/Fr^2 = 1/\varepsilon$ in (8) promotes that same term to $O(\varepsilon)$ in the momentum balance, which is why $\theta_1$ appears in (13). The relative correction in the normalised $\theta$ is still $O(\varepsilon)$, because it is measured against $\theta_0$, whose norm is the largest of the four fields; that normalisation, not a higher order in $\theta_1$, is why $\theta$ has the smallest relative amplitude. The $O(\varepsilon)$ system is linear, with coefficients fixed by $X$. Its mass equation is inhomogeneous, so the operator is not the Boussinesq operator linearised about $X$—whose velocities are divergence-free—but the linearisation of the retained low-Mach system about $X$, with the mass constraint carried inside the operator. We denote that operator by $\mathcal{L}$; its inverse, up to the constant-pressure gauge, is written $\mathcal{L}^{-1}$. It is solvable: its only singular mode is the constant pressure, and the compatibility condition $\int \nabla \cdot (\theta_0 \boldsymbol{u}_0)\, d\Omega = \oint \theta_0 \boldsymbol{u}_0 \cdot \boldsymbol{n}\, dS = 0$ holds because $\boldsymbol{u}_0 \cdot \boldsymbol{n} = 0$ on the walls, so $\tilde{p}_1$ is determined up to the gauge. Since $\mathcal{L}$ is built from $X$ and the retained equations, and contains no explicit $\varepsilon$,

$$D = Y - X = -\varepsilon \mathcal{L}^{-1} S_1[X] - \varepsilon^2 \mathcal{L}^{-1} S_2[X] + \cdots = \varepsilon P(X) + \varepsilon^2 Q(X) + \ldots \tag{14}$$

Five predictions follow: $R \to 0$ as $\varepsilon \to 0$; $R \propto \varepsilon$ with a vanishing intercept; the shape of $D$ does not rotate with $\varepsilon$; $\nabla \cdot \boldsymbol{u}_1 = \nabla \cdot (\theta_0 \boldsymbol{u}_0)$; and the $\varepsilon^1$ term carries most of the correction energy. All five hold; Table S1 of the supplementary material lists them and their measured values. Table S1 also measures the

remainder: the $\varepsilon^1$ term carries 86% of the correction energy, $\varepsilon^1 + \varepsilon^2$ carries 99.0%, and $\varepsilon^1 + \varepsilon^2 + \varepsilon^3$ carries 99.9%, so the two-term truncation leaves 1.0% over the sampled range. This is a measured statement about the retained low-Mach system, not a proof of convergence.

The divergence identity is not exact at the top of the range: the cosine between its two sides falls from 0.994 to 0.901 as $\varepsilon$ rises to 1.9, which is the footprint of the $\varepsilon^2$ term. The fifth row of Table S1 quantifies that term independently.

Domain of validity. $\varepsilon < 2$, since the cold wall $T_c = T_0 - \Delta T/2$ must be positive. At $\varepsilon = 1.9$ the cold wall is $T_c/T_0 = 0.05$, so the density expansion is close to its limit.

## 4. Data sets and protocols

Table S12 summarises the data sets used below: a 600-sample random-boundary scan calibrates the law and trains the surrogates, a $20 \times 5$ paired scan isolates $\varepsilon$, and the public, geometry and fluid sets supply the external checks.

Surrogates use residual learning $\hat{Y} = X + \mathcal{N}(\cdot)$, a U-Net of width 32 (or an FNO2d with 12 modes, width 32, four layers, 1.19 M parameters), Adam with cosine annealing, 200 epochs, batch 16, and targets normalised per channel. Inputs and targets are scaled per channel by standard deviations of the training split; in the cross-data-set transfer experiment the target pool is scaled by the standard deviation of its own inputs, which uses no labels. Two splits are used throughout: extrapolation (train $\varepsilon \leq 1.3$, test $\varepsilon > 1.3$, $n = 218$) and interpolation (stratified 2:1 in $\varepsilon$, $n = 200$). Errors are the per-channel relative $L_2$ error against $Y$; "floor" means $\hat{Y} = X$.

Tables that support the argument without interrupting its flow are collected in the supplementary material and cited where they are used.

Temperature normalisation. Eq. (7) defines the zero-mean variable $\theta = (T - T_0)/\Delta T$. The public release and our own pipeline store $T/T_0$ and evaluate every temperature error on the wall-normalised $\theta_w = (T - T_c)/\Delta T \in [0, 1]$, which differs from $\theta$ by the constant 1/2. All $R_\theta$ and $A_\theta$ quoted below are in the wall-normalised convention. The same field difference reads 0.0220 wall-normalised against 0.0869 zero-mean on the public random-wall test split, and 0.0836 against 0.1628 on the classical cavity; the two conventions must not be mixed. In Table 3 the $\theta$ amplitude is therefore listed in both, 0.010 wall-normalised and 0.034 zero-mean.

## 5. Results

### 5.1. The $\varepsilon$-law, calibrated over two decades

Fig. 1 shows the collapse of $u_x$, $u_y$ and $p$. The $\theta$ panel scatters about the fitted line ($R^2 = 0.22$), because its amplitude also depends on the flow ($\chi$ or $\hat{u}$); Table 3 collects the fitted exponents and amplitudes. The quoted ± is the standard error of the log–log fit. On the 600-sample scan alone the fit gives $m = 1.08$ ($u_x$) and 1.00 ($u_y$); resampling its 134 wall-mode families (2000 cluster bootstraps) gives 95% intervals $[1.03, 1.14]$ and $[0.94, 1.06]$, against $[1.03, 1.14]$ and $[0.95, 1.05]$ under an i.i.d. bootstrap. Clustering by boundary family adds no detectable uncertainty to the exponent, and the shift from 1.08 to 1.02 between the two fits is the effect of the 27 low-contrast points, not of the uncertainty.

Table 3: Power-law fit $R = A\varepsilon^m$ over the full two-decade window (627 samples). The intercept is fitted separately on the small-$\varepsilon$ subset ($\varepsilon \le 0.5$). For $\theta$ the amplitude is given in both conventions, wall-normalised/zero-mean. The $p$ row uses the fixed gauge and hydrostatic correction of Sec. 5.2; its exponent is not gauge-invariant. The $\theta$ row is reported for completeness; its slope is subset-dependent and no exponent is claimed.

| channel | $R = A\varepsilon^m$ | $A$ | $R^2$ | intercept ($\varepsilon \le 0.5$) |
|---|---|---|---|---|
| $u_x$ | $m = 1.023 \pm 0.018$ | 0.147 | 0.84 | $+0.0003 \pm 0.0047$ |
| $u_y$ | $m = 0.995 \pm 0.019$ | 0.193 | 0.82 | $+0.0030 \pm 0.0062$ |
| $\theta$ | $m = 1.024 \pm 0.077$ | 0.010/0.034 | 0.22 | $-0.0003 \pm 0.0012$ |
| $p$ | $m = 1.009 \pm 0.015$ | 0.148 | 0.87 | $+0.0016 \pm 0.0030$ |

The exponent is window-averaged, and the curvature is real. The fit above uses the full two decades, including the 27 check points at $\varepsilon \le 0.1$ where $R$ is closest to its linear limit; these points are in the fit, not a holdout. Fitting only $\varepsilon \ge 0.8$ gives $m = 1.23 \pm 0.08/1.25 \pm 0.08/1.21 \pm 0.34/1.32 \pm 0.06$, and fitting each of the 20 paired configurations separately over $\varepsilon \in [0.4, 1.9]$ gives a median $m = 1.31/1.17/1.21/1.45$ (median $R^2 = 0.98$). The 104 paired configurations used for the Prandtl comparison (Sec. 5.13) give 1.16/1.05/1.67/1.11 over their full range; the $\theta$ slope is the least stable because its signal is the weakest ($R^2 = 0.22$ here and 0.33 in that subset), and it moves much more than the velocity and pressure slopes. That is the signature of $R = A\varepsilon(1 + \kappa\varepsilon)$ rather than of an exact power: a power law fitted to a curved function over a window that excludes small $\varepsilon$ returns $m \approx 1 + \kappa\langle\varepsilon\rangle$. The quadratic fit $R = a\varepsilon + c\varepsilon^2 + b$ gives $c > 0$ in all four channels ($u_x$: $+0.0513 \pm 0.0133$, $3.9\sigma$; $u_y$: $+0.0695$; $\theta$: $+0.0060$; $p$: $+0.0539$) with $\kappa_{\text{fit}} = c/a = 0.70/0.83/0.42/0.73$ (here $a$ and $c$ are coefficients of this quadratic fit, not the power-law $A$ of Table 3). We therefore state the law as first order in $\varepsilon$, $R = A\varepsilon(1 + O(\varepsilon))$, quote $m = 1.02$ as the number a wide-window fit returns, and note that the

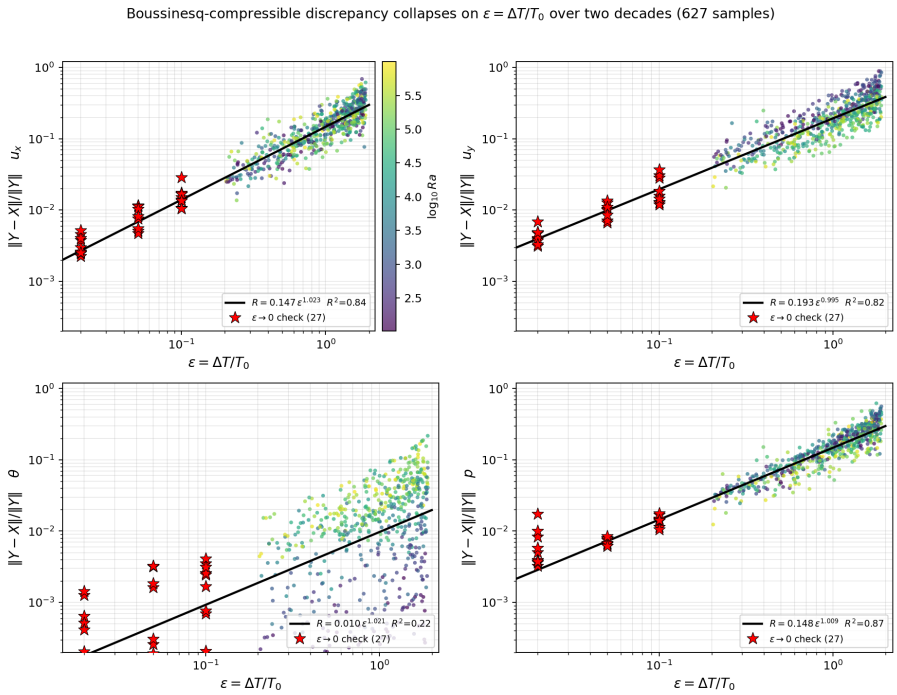



Figure 1: The Boussinesq–compressible discrepancy collapses on $\varepsilon = \Delta T/T_0$ over two decades for $u_x$, $u_y$ and $p$; the $\theta$ panel scatters ($R^2 = 0.22$) and collapses on the peak Boussinesq speed $\hat{u}$ (Fig. S2), while a separate $\chi$ parameterisation gives $R^2 = 0.78$ alone (Sec. 5.1). Points: 600 random-boundary-condition samples, coloured by $\log_{10} Ra$; stars: the 27 small-$\varepsilon$ checks ($\varepsilon = 0.02$–$0.1$, three Rayleigh numbers, three independent wall draws each), included in the fit to anchor the $\varepsilon \to 0$ end and not used as a holdout. Lines: the power-law fits of Table 3.

network recovers the $\varepsilon^2$ curvature with the same sign and order of magnitude (Sec. 5.5, $\kappa_{\text{net}} = 0.22$–$0.42$; P2 fits amplitude and curvature jointly, so $\kappa_{\text{net}}$ is 1.3–2.6 times smaller than $\kappa_{\text{fit}}$ but has the same sign in every channel) and by the mode expansion (Sec. 3, Table S1 of the supplementary material: $\varepsilon^1 + \varepsilon^2$ carry 99.0% of the correction energy).

Mesh independence. One paired configuration (the classical cavity at $Ra = 10^4$) was rerun at $128^2$ over the same five contrasts. The relative correction $R$ changes by at most 0.1% in $u_x$ and $u_y$, 0.7% in $\theta$ and 1.1% in $p$. Fitted over $\varepsilon \in [0.4, 1.9]$, the exponents are 1.192/1.208/1.247/1.305 at $64^2$ and 1.192/1.208/1.247/1.300 at $128^2$ ($u_x/u_y/\theta/p$), so the refinement moves the exponent by at most 0.005. For this configuration the exponent is not set by the $64^2$ discretisation.

At the top of the Rayleigh range we repeated the check for a constant-wall configuration at $Ra = 10^6$, over $\varepsilon = 0.4$, 1.2 and 1.9. The exponents are 1.067, 1.129, 1.301 and 1.335 at $64^2$, against 1.063, 1.124, 1.289 and 1.321 at $128^2$ (for $u_x$, $u_y$, $\theta$ and $p$), a shift of at most 0.015. The largest pointwise change in $R$ is 3.3% ($p$ at $\varepsilon = 1.9$); the other channels change by 1.0–1.9%. The thermal boundary layer at $Ra = 10^6$ is $\approx 0.032L$ thick; the $64^2$ mesh places three temperature and five velocity nodes across it, the $128^2$ mesh five and nine. The $128^2$ values are the reference for the exponent and pointwise checks: the $64^2$ exponent is accurate to 0.015, and an individual $\varepsilon = 1.9$ correction carries a discretisation uncertainty of up to 3.3%. A $192^2$ pair at $\varepsilon = 1.9$ and a $256^2$ pair at $\varepsilon = 1.2$ set the level of the integral correction.

The $Ra = 10^5$ check, the $192^2$ and $256^2$ refinements, and the $\varepsilon = 1.9$ $256^2$ run that did not reach tolerance are reported in Supplementary Sec. S10.

- For $u_x$, $u_y$ and $p$ the exponent is consistent across geometry within its uncertainty (0.7–1.8 standard errors; $p$ in the fixed gauge); the prefactor $A$ is not ($0.147 \to 0.175$ for $u_x$, $0.148 \to 0.222$ for $p$, $0.193 \to 0.175$ for $u_y$, $0.010 \to 0.008$ for $\theta$).

- Boundary-condition families barely matter inside this family: 134 wall-mode families, 133 free dummies. Adding them to the $\varepsilon$-only fit raises the adjusted $R^2$ by 2.3 pp ($p$), 2.9 pp ($u_y$), 3.3 pp ($u_x$) and 8.5 pp ($\theta$)—the $\varepsilon$-scaling is not an artefact of which wall modes were drawn.

- The $\theta$ channel needs one extra, cheaply available parameter, the advection strength of the base flow $\chi = \|\boldsymbol{u}\|\|\nabla\theta\|/\|\theta\|$: $R_\theta = \varepsilon(-0.0037 + 0.226\chi)$ with the $\varepsilon$ factor imposed, not fitted; $R^2$ 0.185 $\to$ 0.778 (Spearman 0.902), i.e. $R_\theta \approx 0.23\varepsilon\chi$. This is a parameterisation of the $\theta$ amplitude, not an independent measurement of its exponent. For $u_x/u_y/p$ adding $\chi$ does nothing. A single-input form of the same statement, $R_\theta/\varepsilon = c\hat{u}^a$, is calibrated on two independent generators in Sec. 5.7 and carries the same information with $R^2 = 0.90$.

- The fitted $\theta$ amplitude is not predictive: $A_\theta = 0.010$ with $m = 1.024$ gives 0.012 at $\varepsilon = 1.2$, while the median $R_\theta$ on the public data is 0.0228. The factor 1.9 is the 78% unexplained variance of the $\theta$ fit, and is a further reason no $\theta$ exponent is claimed.

- The pressure channel required a gauge and hydrostatic correction before the two members could be compared (Sec. 5.2).

5.2. The pressure channel: gauge and hydrostatic alignment

The compressible member stores a hydrostatic ramp 12× larger than its dynamic pressure and the two weak forms use opposite signs, so raw

pressures measure the conventions rather than the flow ($R_p \approx 1.0$). After subtracting the mean, removing a least-squares linear-in-$y$ fit and flipping the sign of the Boussinesq member, the aligned pair matches the public data set to 3.5% and follows the velocity law ($A = 0.148$, $m = 1.009$). All $p$-channel numbers use this fixed gauge and are therefore not gauge-invariant. Supplementary Sec. S4 gives the raw and aligned statistics and the protocol.

### 5.3. The $\varepsilon$-blind ceiling, measured exactly

Within a configuration $X$ is identical for all $\varepsilon$, so an $\varepsilon$-blind model must emit one field $F$ for every $\varepsilon$ it is applied at (Fig. 2, Table 4). Two reference protocols are computed directly from the paired data rather than fitted: (A) train at the one $\varepsilon$ a publicly released data set offers (here $\varepsilon = 1.2$) and apply it everywhere, error $\|Y(\varepsilon) - Y(1.2)\|/\|Y(\varepsilon)\|$; (B) one field chosen without $\varepsilon$, the unweighted $\varepsilon$-mean of $Y$. (B) is a compromise fixed by that choice of average, not a per-$\varepsilon$ lower bound—a field tuned to $\varepsilon = 0.8$ gives 0.067 at $\varepsilon = 0.4$, better than (B)'s 0.180 there, and below $\varepsilon \approx 0.8$ doing nothing beats both. Median over 20 configurations, channels $u_x/u_y/\theta/p$:

Table 4: An $\varepsilon$-blind reference. Protocol (A) is what a single-$\varepsilon$ public data set forces; protocol (B) is the unweighted $\varepsilon$-mean field, a reference rather than an optimum for the relative $L_2$ error. Entries are medians over 20 configurations, ordered $u_x/u_y/\theta/p$; every entry is a relative $L_2$ error with $\|Y\|$ in the denominator, the same convention as the surrogate errors.

| $\varepsilon$ | do nothing ($\hat{Y} = X$) | (A) train at $\varepsilon = 1.2$ only | (B) $\varepsilon$-mean field |
|---|---|---|---|
| 0.4 | 0.060/0.098/0.009/0.060 | 0.149/0.229/0.021/0.159 | 0.180/0.255/0.023/0.196 |
| 0.8 | 0.124/0.198/0.019/0.123 | 0.081/0.121/0.011/0.086 | 0.107/0.141/0.013/0.134 |
| 1.2 | 0.200/0.307/0.031/0.201 | 0/0/0/0 | 0.035/0.036/0.003/0.053 |
| 1.6 | 0.292/0.440/0.045/0.298 | 0.110/0.144/0.014/0.121 | 0.077/0.117/0.012/0.080 |
| 1.9 | 0.388/0.575/0.060/0.424 | 0.243/0.295/0.031/0.273 | 0.209/0.268/0.028/0.235 |

Protocol (A) is what a single-$\varepsilon$ public data set forces: 24% ($u_x$), 30% ($u_y$), 3.1% ($\theta$) and 27% ($p$) error at $\varepsilon = 1.9$, 8% ($u_x$) already at $\varepsilon = 0.8$, and—in practice—at $\varepsilon = 0.4$ "do nothing" is better than it, and the two cross between $\varepsilon = 0.4$ and 0.8. Protocol (B), the unweighted mean over the range, still leaves 20.9/26.8/2.8/23.5% ($u_x/u_y/\theta/p$) at $\varepsilon = 1.9$. The effect is algebraic, and the paired set measures it exactly (the diagonal is identically zero because the same $\varepsilon$ is the same solution): the full $5 \times 5$ matrix for $u_x$ is Table S3 of the supplementary material.

Within a paired configuration the error is set by the distance in $\varepsilon$—which is precisely the quantity $\varepsilon P(X)$ puts back into the model.

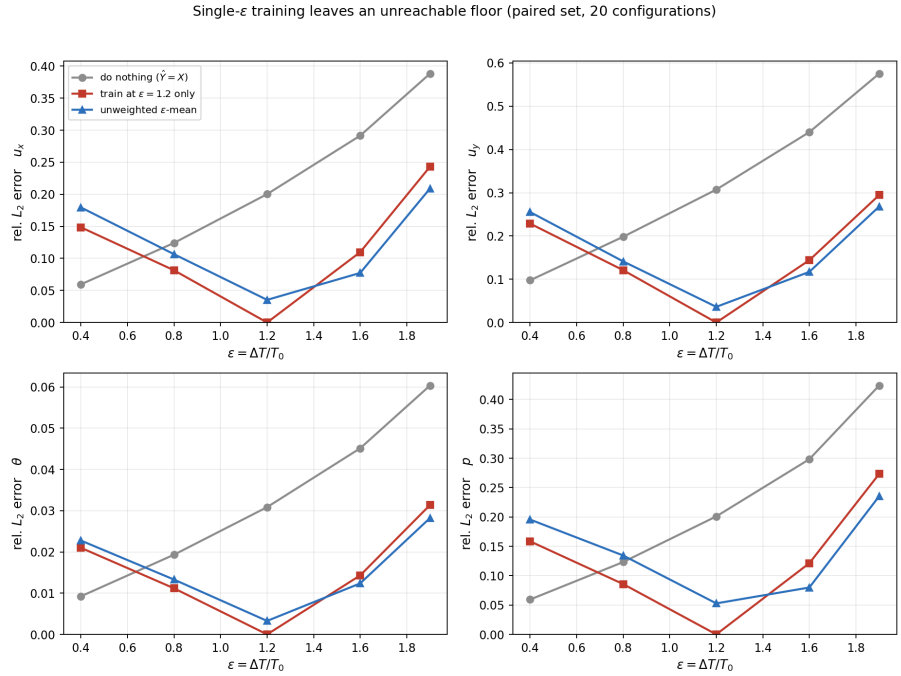



Figure 2: Single-$\varepsilon$ training leaves an $\varepsilon$-blind floor. Curves show "do nothing", training at $\varepsilon = 1.2$ only, and the unweighted $\varepsilon$-mean field, for the 20 paired configurations.

The shape of the correction is stable even though its amplitude is not. Over the 20 paired configurations, the median cosine similarity between $D(\varepsilon)$ and $D(1.2)$ stays above 0.952 over $\varepsilon = 0.4$–$1.9$, with minima 0.974 ($u_x$), 0.976 ($u_y$), 0.982 ($\theta$) and 0.952 ($p$; Table S11). The largest deviation from unity is 0.048; a scalar amplitude therefore carries the contrast dependence, as Eq. (14) assumes.

### 5.4. Network baselines: the missing input is a scalar, not capacity

Table 5 reports a U-Net of width 32, 200 epochs, mean of three seeds, identical protocol for all columns (largest standard deviation over every cell: 0.0051).

Three seeds give the resolution. The paired B−P1 difference is positive in every channel in both splits: extrapolation 0.057/0.067/0.011/0.067 and interpolation 0.013/0.019/0.002/0.021 ($u_x/u_y/\theta/p$), and every 95% interval excludes zero (half-widths 0.001–0.020). The FiLM comparison is weaker: in extrapolation the mean differences are 0.006/0.010/0.002/0.012, but the intervals for $u_x$ (−0.001 to 0.013) and $u_y$ (−0.009 to 0.028) include zero; in interpolation the differences are −0.000/−0.003/−0.001/−0.003 and all intervals include zero. We therefore claim an advantage over FiLM for the

Table 5: Surrogate ablation on the two fixed splits (U-Net, width 32, 200 epochs, mean of three seeds; largest standard deviation over all cells 0.0051). A sees only the four Boussinesq fields; B also receives $\varepsilon$ as a constant input channel; P1 multiplies $\varepsilon$ outside the network but still reads it as an input; P2 adds a learned quadratic factor; P3 removes $\varepsilon$ from the input entirely. Errors are per-channel relative $L_2$ errors against $Y$. The curvature learned by P2 (seed 0) is $\kappa_{\text{net}} = 0.418/0.314/0.334/0.314$ (extrapolation) and $0.295/0.224/0.255/0.219$ (interpolation), for $u_x/u_y/\theta/p$; the three-seed means are $0.418/0.321/0.343/0.338$ and $0.286/0.232/0.252/0.265$. The own-data row of Table 9 is the P3 architecture trained with 150 epochs and one seed, the protocol of the public-transfer rows; this table uses 200 epochs and three seeds; the same architecture therefore differs between the tables by 2–6% in extrapolation and 6–12% in interpolation.

| split | channel | A | B | P1 | P2 | P3 |
|---|---|---|---|---|---|---|
| extrapolation ($n = 218$) | $u_x$ | 0.2289 | 0.1508 | 0.0937 | 0.0867 | 0.1214 |
| | $u_y$ | 0.2733 | 0.1725 | 0.1060 | 0.0951 | 0.1277 |
| | $\theta$ | 0.0406 | 0.0255 | 0.0141 | 0.0131 | 0.0161 |
| | $p$ | 0.2689 | 0.1791 | 0.1126 | 0.1007 | 0.1682 |
| interpolation ($n = 200$) | $u_x$ | 0.1530 | 0.0673 | 0.0545 | 0.0507 | 0.0732 |
| | $u_y$ | 0.1796 | 0.0714 | 0.0524 | 0.0488 | 0.0749 |
| | $\theta$ | 0.0261 | 0.0097 | 0.0078 | 0.0073 | 0.0092 |
| | $p$ | 0.1792 | 0.0667 | 0.0454 | 0.0410 | 0.0893 |

extrapolation mean, not for individual velocity channels and not in interpolation.

Model A's error on the extrapolation split (0.2289) is 9% above the $\varepsilon$-mean reference on the paired set at $\varepsilon = 1.9$ (0.2091, Sec. 5.3); the two are measured on different data sets and with different training sets, but their agreement in magnitude is consistent with a ceiling that is informational rather than architectural. Shallow learners plateau in the same place. A linear ridge regression and an RBF kernel ridge on the raw four-field input—the same information the network gets—reach $u_x$ 0.2282 and 0.2222 in extrapolation, and adding $\varepsilon$ as an extra column barely moves them (0.2188 and 0.2171, against 0.0937 for the structured model), because neither can represent a product of an $\varepsilon$-scale with a spatial field. A classical POD-ROM (44–52 modes, 99.9% of the snapshot energy) whose coefficients are regressed on $(\log_{10} Ra, \varepsilon)$ lands at 0.2937 with $\varepsilon$ and 0.3001 without, i.e. at the level of doing nothing, and a purely linear surrogate that also sees the POD coefficients of $X$ reaches only 0.2318. The plateau is not capacity—it is that the factor scaling the quantity to be added is absent from the input.

A first-order perturbation proxy. If the correction is linear in $\varepsilon$, the alternative is to solve the linearised compressible problem once per query and scale. That costs one Jacobian assembly and one factorisation of the same discretisation. Measured on the linearised compressible system at $Ra = 10^6$

(45 957 degrees of freedom), one solve takes 1.9 s: 0.12 s assembly, 1.59 s LU factorisation and 0.02 s back-substitution, the remaining 0.17 s being mesh construction and state input/output. Both $P$ and the network are identified offline; the comparison here is per query. The field that solve produces is $P = \lim_{\varepsilon\to 0} D(\varepsilon)/\varepsilon$, the solution of $\mathcal{L}\delta = S_1[X]$. We first estimate $P$ without a new solve: the paired scan already contains $D(\varepsilon)$, so we estimate $P$ by third-order Richardson extrapolation in $\varepsilon$ over the solutions at $\varepsilon = 0.4, 0.8, 1.2$. With $\mathcal{P}(\varepsilon) = D(\varepsilon)/\varepsilon$, $P = 3\mathcal{P}(0.4) - 3\mathcal{P}(0.8) + \mathcal{P}(1.2)$, which has $O(\varepsilon^3)$ truncation and the same norm as the $\varepsilon = 0.4$ secant to within 1% in the median over the 20 configurations, in all four channels. The accuracy below is that of this extrapolated field. At 6.3 ms the network is about 300 times cheaper per query than this linear solve, before any allowance for the different hardware. Predicting $\hat{Y} = X + \varepsilon P$ gives 14/15/1.4/15% at $\varepsilon = 1.6$ and 24/27/3.0/28% at $\varepsilon = 1.9$ ($u_x/u_y/\theta/p$), against a do-nothing floor at $\varepsilon = 1.9$ of 39/58/6.0/42%: the linear response removes 34–53% of the gap at $\varepsilon = 1.9$ (computed from the unrounded errors; $p$ gives 34.9%) and leaves the rest to the $O(\varepsilon^2)$ term. The secant at $\varepsilon = 0.4$ predicts 22/26/2.9/26% at $\varepsilon = 1.9$, slightly better than the linear response itself, because the secant already contains part of the quadratic term; it costs one full compressible solve (30.3 s median on this 20-configuration scan; 22.3 s is the 600-sample median reported in Table S10) and using it as the linear baseline would flatter that baseline. The network reaches 9.4/10.6/1.4/11.3% on the extrapolation split (Table 5), so the residual left by the linear response is not a small correction. Both approaches replace the same compressible member and therefore save a similar fraction of the 30.3 s nonlinear solve; the first-order solve cannot produce the $\varepsilon^2$ term, while inference costs 6.3 ms against 1.9 s per linear solve.

Direct solution of the $O(\varepsilon)$ system. The extrapolated $P$ above is fitted to the paired data, so it needs a check that is not another fit. Let $F(U;\varepsilon)$ be the steady residual of (1)–(4). One Newton step from the Boussinesq state, $\delta = -J^{-1}F(X)$ with $J = \partial F/\partial U$ evaluated at $X$, solves the linearised system that defines $P$: because $F(X) = O(\varepsilon)$, $\delta = D + O(\varepsilon^2)$ and therefore $\delta/\varepsilon = P + O(\varepsilon)$. Measured on the 27 samples of the low-contrast sweep ($\varepsilon = 0.02, 0.05, 0.10$; $Ra = 10^3$–$10^5$; three independent wall draws each), where the correction is 0.4–5.8% of $X$, $\delta/\varepsilon$ reproduces the measured $D/\varepsilon$ with median relative errors of 4.7%, 3.8%, 7.2% and 12.2% in $u_x$, $u_y$, $p$ and $\theta$ (10–90% ranges 0.8–19.6%, 0.5–20.8%, 0.8–39.2%, 1.4–24.6%) and median correlations 0.999, 0.999, 0.998 and 0.994. No extrapolation in $\varepsilon$ and no fitted $P$ enter this comparison. The same comparison on the paired scan gives 15% in $u_x$ at $\varepsilon = 0.4$ and 73% at $\varepsilon = 1.9$, rising linearly in $\varepsilon$: a single Newton step is an $O(\varepsilon)$ instrument, and what it leaves behind at high contrast is the $O(\varepsilon^2)$ term. At fixed flow, $\delta/\varepsilon$ itself moves by 12.5% in $u_x$ between $\varepsilon = 0.1$ and 0.4 and by 153% between 0.4 and 1.9 (median over 17 configurations), while the measured $D/\varepsilon$ moves by 9.0% and 102%: the residual and its Jacobian

carry $\varepsilon$ explicitly through $1/Fr^2 = 1/\varepsilon$, so a solve at finite contrast is not yet the limit and the extrapolation is doing real work. The extrapolated $P$ is then tested against data it never saw. Nineteen of the paired configurations were re-solved at $\varepsilon = 0.1$ and 0.2 from the same random stream, giving 38 solutions that enter no fit; the test holds out the contrast, not the flow. There $P$ predicts the measured $D/\varepsilon$ with median relative errors of 2.4%, 1.8% and 3.8% ($u_x/u_y/p$); the $\theta$ channel is not predicted (74%), consistent with its unidentifiable exponent. Because $D(0.1)/0.1$ and $D(0.2)/0.2$ differ by only 3.1% in $u_x$, the three-point combination is estimating a limit that the data approach, not a local secant. Both checks use the same mixed space as the reference solver and one LU factorisation per configuration.

### 5.5. The physics prior: $D = \varepsilon P(X)$

Same backbone, only the output structure changes (Table 5 of Sec. 5.4). B keeps the target $D$ and reads $\varepsilon$ from a constant input channel; P1 reads $\varepsilon$ as well but trains on $D/\varepsilon$ and multiplies $\varepsilon$ back outside; P3 removes the $\varepsilon$ input entirely.

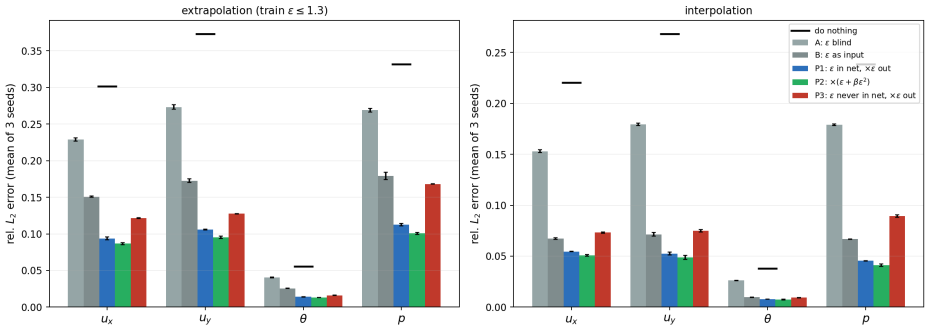



Figure 3: The prior ablation for extrapolation (left) and interpolation (right). A: $\varepsilon$-blind. B: $\varepsilon$ as input. P1: $\varepsilon$ in the network, multiplied outside. P2: learned $(\varepsilon + \kappa\varepsilon^2)$ factor. P3: $\varepsilon$ never enters the network.

Three things happen at once (Table 5, Fig. 3). In extrapolation the factorised output beats a plain network that reads $\varepsilon$ from a constant input channel by 1.61× ($u_x$) and 1.59–1.81× in the other channels, and the quadratic form adds a further 7.1–10.6% on top. That channel is a weak way to condition a network. A FiLM-conditioned U-Net, which modulates every block with $\varepsilon$, reaches 0.1001/0.1155/0.0164/0.1251 in extrapolation and 0.0544/0.0496/0.0073/0.0428 in interpolation, so against that baseline the channel-averaged error of the factorised output is 1.09× lower in extrapolation (1.16× in $\theta$ and 1.11× in $p$ in the fixed gauge; the $u_x$ and $u_y$ intervals include zero), while the two are indistinguishable in interpolation. Removing $\varepsilon$ from the input as well costs accuracy relative to P1 and FiLM—0.1214 against 0.0937 and 0.1001 in $u_x$—but still beats the $\varepsilon$-blind network (1.87×

channel-averaged on the same 600-sample scan; 1.89× in $u_x$); what it buys is the property P1 cannot have, a network that can be trained at one $\varepsilon$ and applied at another (Sec. 5.9). Finally, the curvature learned by P2 is positive, $\kappa_{\mathrm{net}} = 0.22$–$0.42$, the same sign and order as the $\varepsilon^2$ term of the expansion in Sec. 3—an independent check of the asymptotic expansion.

Practical form. The prior changes only the training target: train any regressor $\mathcal{N}$ on $D/\varepsilon$ instead of $D$, and at run time return $\hat{Y} = X + \varepsilon\,\mathcal{N}(X)$ after evaluating the Boussinesq member (or a surrogate for it) to obtain $X$. For wall heat transfer, add $\lambda \sum |\hat{q} - q| / \sum |q|$ to the loss, where $q$ is the wall-normal gradient of $\theta$; $\lambda = 1$ with the six-point stencil is enough (Sec. 5.8). Supplying $\varepsilon$ as well (P1) is the most accurate form; leaving it out (P3) costs accuracy at fixed data but lets one network serve every contrast for which the prefactor is unchanged.

What a single-contrast data set can and cannot identify. Training on $D/\varepsilon_0$ at one contrast returns $P + \varepsilon_0 Q + O(\varepsilon_0^2)$, not $P$; applied at another contrast the leading term carries the error $\varepsilon(\varepsilon_0 - \varepsilon)Q$. Over the paired scan $|Q|/|P| = 0.56$ in $u_x$, 0.31 in $u_y$ and 0.79 in $p$; for $\varepsilon_0 = 1.2$ applied at 0.4 the error in $u_x$ is $0.18|P|$ against a correction of $0.4|P|$. The two amendments that help both act on the amplitude: the curvature term of P2 and $\varepsilon$ as an input (P1). A single-contrast data set can only supply $P + \varepsilon_0 Q$.

A Fourier neural operator (1.19 M parameters) shows the same pattern against its $\varepsilon$-blind version: $u_x$ 0.2287 → 0.1334 with the prior and $\varepsilon$ multiplied back outside the network (1.71×), statistically indistinguishable from an explicit $\varepsilon$ input channel ($0.1334 \pm 0.0004$ against $0.1342 \pm 0.0008$). On $u_y$ and $\theta$ the prior is in fact better (0.1387 against 0.1456; 0.0177 against 0.0191); only on $p$ does the $\varepsilon$ channel win (0.1536 against 0.1734). The prior's value is therefore not that it beats an $\varepsilon$-aware model—it is that it matches one while needing no $\varepsilon$ in the input (Secs. 5.6, 5.9).

### 5.6. Data efficiency

The learning curve is tabulated in Table S4 of the supplementary material and plotted in Fig. 4.

The naive expectation—that a structured model trades in-distribution accuracy for extrapolation—is wrong at small $n$. At $n = 50$ the prior is 1.4× better than the $\varepsilon$-input network ($u_x$ 0.1308 against 0.1831, a $9\sigma$ separation across three seeds), and it is still ahead at $n = 200$ in $u_x$ ($7.3\sigma$), $u_y$ ($7.6\sigma$) and $\theta$ ($3.6\sigma$), with $p$ already level ($0.13\sigma$) there. The crossover sits between $n = 200$ and 400 in $u_x$ and $u_y$ (log-linear interpolation: $n \approx 315$ and 360, i.e. near 300 in $u_x$), essentially at $n = 200$ in $p$, and $\theta$ never crosses—at $n = 400$ it is still $2.5\sigma$ behind. Concretely, the prior at $n = 200$ reaches what the $\varepsilon$-input network needs $\approx 270$ samples ($u_x$) and 290 ($u_y$) to reach. So the

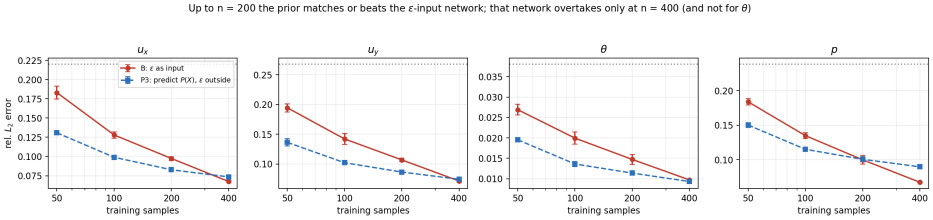



Figure 4: Learning curves. Up to $n = 200$ the prior matches or beats the $\varepsilon$-input network; the latter overtakes only at $n = 400$, and not for $\theta$.

prior is two things at once: the cheaper model when data is scarce, and—a model that needs no $\varepsilon$ in its input, which is what makes a single-$\varepsilon$ data set reusable.

### 5.7. External validation on an independent, publicly released data set

The public release NeuralConvection_2D_TrainTest (v2; 4000 training pairs plus three 200-pair held-out splits, CC BY 4.0) was generated by another group, at a single $\varepsilon = 1.2$, $Ra \in [10^2, 3 \cdot 10^6]$ [5].

Convention conversion. Recomputing their published relL2_* columns as $\|Y-X\|/\|X\|$ from the tensors reproduces them exactly (0.14233/0.13749/0.01356/0.19070 for sample 0); their pressure amplitude ratio is 0.929 against 0.960 for our fixed pairing, under the same median convention. The comparison below uses $\|Y - X\|/\|Y\|$, the convention of the rest of this paper; the published columns use $\|X\|$ and are reproduced to round-off before conversion.

The law reproduces. Table 6 compares the $\varepsilon = 1.2$ slice of the public release with our own data at the same contrast.

Table 6: External validation at $\varepsilon = 1.2$: public data against our own slice. Medians with p10–p90 in parentheses.

| channel | public 4000 (median, p10–p90) | our data at $\varepsilon \in [1.1, 1.3]$ ($n = 64$) |
|---|---|---|
| $u_x$ | 0.1797 (0.112–0.300) | 0.1839 (0.126–0.284) |
| $u_y$ | 0.2435 (0.141–0.403) | 0.2250 (0.150–0.395) |
| $\theta$ | 0.0228 (0.001–0.071) | 0.0232 (0.001–0.056) |
| $p$ | 0.1858 (0.117–0.246) | 0.1889 (0.128–0.262) |

Even the secondary trends reproduce. The correlation of $R$ with $\log_{10} Ra$ has the same signs on the two data sets; the $\theta$ channel, whose scatter we explain next, has $r = +0.66$ on ours and $+0.65$ on theirs, from independently generated data.

Five splits, four prefactors, one predictable pattern. The release publishes three 200-pair held-out splits (in-distribution, out-of-distribution wall dials, and the classical cavity) plus 5000 classical-cavity training pairs, all at $\varepsilon = 1.2$. Recomputing all 38 400 published error values under our conventions reproduces them to $1.2 \times 10^{-7}$, so nothing below is a convention artefact; the per-split medians are in Table S5 and the split definitions in Supplementary Sec. S8.

The exponent cannot be re-tested here—every split sits at one $\varepsilon$—but the prefactor is put under pressure and it moves: by 1.66× ($u_x$), 1.38× ($u_y$), 4.05× ($\theta$) and 1.07× ($p$) between the classical cavity and the random-wall families at the same $\varepsilon$, and train_legacy and test_legacy agree to three digits in $u_x$ and $p$ ($\theta$: 0.0926 vs 0.0836, 10%), which is the internal consistency check for the whole comparison. That is a statement about the flow, not a defect of the law, and the $\theta$ channel makes the mechanism explicit. Writing $\hat{u} = \max \|\boldsymbol{u}_X\|$ for the peak speed of the Boussinesq field, which is available to any surrogate, the $\theta$-channel prefactor collapses on $\hat{u}$ (Fig. S2); the fits are in Table S6 of the supplementary material.

Two independent groups, two independent generators, six data sets: the random-wall families all give $c = 0.54$–$0.69$, the classical cavity—whose flow is a single strong circulation with $\hat{u}$ 41% above the public random-wall median and 56% above ours—gives $c = 2.06$–$2.10$ . The same reading explains the residual scatter in $u_x/u_y/p$ in Sec. 5.1: $A$ is a functional of the flow, and the flow is in the input.

### 5.8. What the correction does to the wall heat transfer

Every error quoted so far is a field error. For a heat-transfer application the quantity that matters is the Nusselt number, and the public classical cavity answers the question directly at $\varepsilon = 1.2$, because the two members share mesh and boundary conditions. We compute $Nu = -\langle \partial\theta/\partial n \rangle$ at the hot wall with a six-point one-sided stencil. On the Boussinesq member the same flux can be obtained without any wall stencil, from the horizontally integrated flux $\int [u_x\theta - (Ra\,Pr)^{-1/2}\partial_x\theta]\,dy$, which is independent of $x$ to 0.1% at the three interior planes $x = 0.25, 0.5, 0.75$; the wall estimate agrees with it to 0.35%, which indicates the discretisation error of the estimator; the integral change reported below is 1.5 times that error.

FiLM is not a separate row here: the public data have a single $\varepsilon$, so its conditioning input is constant and adds no information beyond the plain model. The FiLM comparison across $\varepsilon$ is the field-error comparison of Table 5.

The integral heat transfer is insensitive: the compressible member carries more heat in all 200 samples, by 0.01–0.94% (median 0.58%); +0.54% is the mean of the per-sample ratios at the six-point stencil, +0.67% the ratio of the ensemble means, and the stencil range moves the former by 0.35–0.62%.

Table 7: Wall heat transfer of the public-data surrogate at $\varepsilon = 1.2$ (200 classical-cavity samples). The surrogate is trained on the 4000 public random-wall training pairs and evaluated, without retraining, on the 200 classical-cavity held-out pairs—a different boundary-condition family at the same $\varepsilon$. $T$ error is relative $L_2$; $Nu$ error is the mean absolute error of the integrated wall Nusselt number divided by the mean true $Nu$; local flux is the mean absolute error of the wall-normal flux profile divided by its mean absolute value. For doing nothing the $Nu$ error is 0.67% of the ensemble-mean $Nu$; the mean of the per-sample ratios is +0.54% (Table S13), and all 200 per-sample changes are positive. The last row adds a relative $L_1$ penalty on the wall-normal gradient to the training loss (weight 1, two seeds).

| model | $T$ error | $Nu$ error (hot) | $Nu$ error (cold) | local flux (hot/cold) |
|---|---|---|---|---|
| do nothing ($\hat{Y} = X$) | 7.2% | 0.67% | 0.64% | 6.0/5.2% |
| plain $D$ | 0.10% | 3.61% | 2.42% | 4.4/3.7% |
| prior $D = \varepsilon P(X)$ | 0.10% | 2.56% | 1.72% | 4.3/3.6% |
| prior + wall-flux loss | 0.13% | 0.11% | 0.10% | 1.0/0.9% |

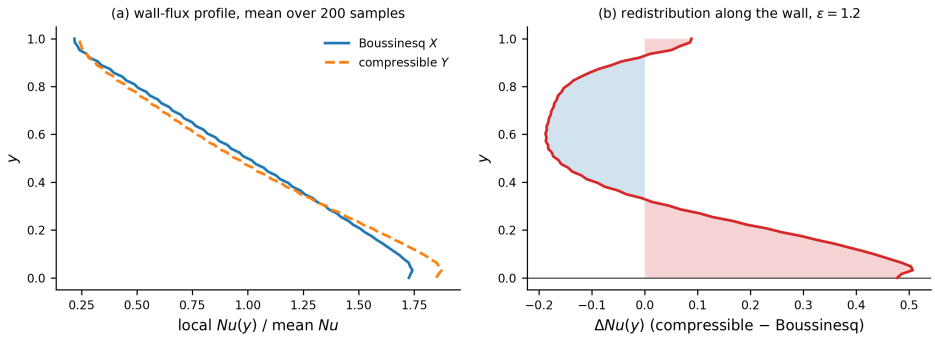



Figure 5: Wall heat transfer of the pair at $\varepsilon = 1.2$. (a) Local Nusselt number along the hot wall, normalised by its mean, for the two members. (b) Redistribution along the wall: the compressible member carries more heat in the lower half and less in the upper half, while the integral changes by less than 1%.

The local wall flux is not: the correction redistributes it by 4.25% in $L_1$ (p90 7.33%) with pointwise excursions of ±10%, adding heat over the lower half and removing it over the upper half (Fig. 5b); the integral is the $Nu$-weighted mean of this redistribution, which is why it cancels. Doing nothing predicts the integral heat load to 0.7% only because it predicts no change at all. Table 7 evaluates the public-data prior directly: the temperature error falls from 7.2% to 0.10% and the local wall-flux $L_1$ error from 6.0% to 4.3% (hot) and 5.2% to 3.6% (cold), but the integrated wall-$Nu$ error is 2.6%, larger than the 0.7% of doing nothing. A relative $L_1$ penalty on the wall-normal gradient removes it: 0.11% integrated and 1.0% local, at 0.10% → 0.13% in the field norm; evaluated with four- and eight-point stencils the same models give 0.3–0.9% against 1.8–3.6% for the field-only prior. The absolute

0.11% is below the 0.35% discretisation uncertainty of the wall estimator itself: the audit establishes the ordering—a field-only target leaves a 2.6% bias that the penalty removes—not the absolute value. The size of the gain depends on the stencil, its sign does not.

Transfer of the wall-flux penalty across contrast. The penalty above is trained at $\varepsilon = 1.2$. We apply the same weights, without retraining, to our 600 random-wall samples at the other contrasts, and keep only samples with $|Nu_Y| > 1$ at the left wall, where the relative error has a non-degenerate denominator (24, 46, 55 and 11 samples at $\varepsilon = 0.4$, 0.8, 1.6 and 1.9). The integral-$Nu$ error is 0.17, 0.20 and 1.21% at $\varepsilon = 0.4$, 0.8 and 1.6, against 0.67, 1.08 and 2.30% for the same prior without the penalty and 0.45, 1.08 and 4.08% for doing nothing (Table 8). The paired improvement over the prior is 0.49, 0.88 and 1.09 percentage points, with 95% bootstrap intervals that exclude zero. At $\varepsilon = 1.9$ the penalty gives 2.92% against 2.66% for the prior and 3.44% for doing nothing; with eleven samples the paired difference is not resolved ($[-1.4, +2.2]$ percentage points). The penalty therefore transfers over the range where the correction is still small enough for the wall gradient to be resolvable; at the largest contrast it does not improve on the field-only prior.

Table 8: Integral-$Nu$ error at unseen contrasts for the public-data models trained at $\varepsilon = 1.2$. Errors are $|Nu_{\text{pred}} - Nu_Y|/|Nu_Y|$ at the left wall, averaged over the samples with $|Nu_Y| > 1$; $n$ is the number of such samples. The last column is the paired difference (penalty minus prior), point estimate and 95% bootstrap interval.

| $\varepsilon$ | $n$ | do nothing | prior | prior + wall flux | paired $\Delta$ (pp) |
|---|---|---|---|---|---|
| 0.4 | 24 | 0.45 | 0.67 | 0.17 | $-0.49$ $[-0.71, -0.30]$ |
| 0.8 | 46 | 1.08 | 1.08 | 0.20 | $-0.88$ $[-1.25, -0.59]$ |
| 1.6 | 55 | 4.08 | 2.30 | 1.21 | $-1.09$ $[-1.57, -0.62]$ |
| 1.9 | 11 | 3.44 | 2.66 | 2.92 | $+0.26$ $[-1.43, +2.20]$ |

The field statistic behind the cancellation is large: on the same 200 samples the temperature field differs by 7.2% in global relative $L_2$ (8.4% median per sample), so a 7–8% change in the field produces a 0.6% change in its wall-normal derivative.

The relative gap is not a meaningful statistic for every configuration. Of the 20 paired configurations, eight have $|Nu_X| \leq 1$ at $\varepsilon = 1.9$ and nine have negative $Nu_X$; there the sign or the small magnitude of the denominator, not the correction, sets $\Delta Nu/Nu$, and the all-configuration mean +5.06% is not a physical heat-transfer result. The five isothermal-sidewall configurations have $Nu_X = O(1)$ at every contrast. This family is the classical cavity, fixed before the analysis; it is not selected by the outcome. On those, the mean integral change is +0.51% (sd 0.37%) at $\varepsilon = 1.2$ and −1.89% (sd 1.94%) at $\varepsilon = 1.9$. The point estimate changes sign, but with five configurations

the 95% interval of the $\varepsilon = 1.9$ mean is $[-4.3, +0.5]$% and includes zero; we report the sign change as a descriptive trend consistent with the $\varepsilon^2$ term, not as a significant effect. A Boussinesq-trained surrogate cannot produce it because it has no access to the compressible correction.

The Rayleigh-number dependence of the integral correction. The five Rayleigh numbers span +0.01 to +0.87% at $\varepsilon = 1.2$ and −5.1 to −0.01% at $\varepsilon = 1.9$ (Table S14). At $Ra = 10^2$ the change is at or below 0.01%; at $Ra = 10^3$ it stays within 1%; at $Ra \geq 10^4$ it reaches −5%. The 15 random-wall configurations are excluded: eight have $|Nu_X| \leq 1$ at $\varepsilon = 1.9$, where the sign or the small magnitude of the denominator, not the correction, sets the ratio. A model calibrated on $Nu$ alone therefore does not transfer across $Ra$ at fixed $\varepsilon$; the local field is the quantity that carries the correction.

### 5.9. Cross-data-set transfer: the payoff, and its boundary

Two public single-$\varepsilon$ data sets are available (Table 9, Fig. 6): 4000 random-wall pairs (our boundary-condition family) and 5000 classical-cavity pairs (a different one). Both sit at $\varepsilon \equiv 1.2$. We train the same U-Net on each—once with the plain output (target $D$), once with the prior (target $D/\varepsilon$, and $\varepsilon$ multiplied back outside the network)—and apply the result to our own 600 samples with no retraining, at $\varepsilon$ from 0.2 to 1.9, on both sides of the single training contrast.

Table 9: Transfer from a single-$\varepsilon$ public data set to our own $\varepsilon \in [0.2, 1.9]$ samples, on the extrapolation and interpolation test splits. Entries are $u_x/u_y/\theta/p$. Both public sets are at $\varepsilon \equiv 1.2$. The "our own training split" row uses 382 training samples for the extrapolation column and 400 for the interpolation one, with a prior that has no $\varepsilon$ input (150 epochs, one seed), the same protocol as the public rows above; the P1/P3 columns of Table 5 use 200 epochs and three seeds.

| trained on | model | extrapolation ($n=218$) | interpolation ($n=200$) |
|---|---|---|---|
| do nothing | $\hat{Y} = X$ | 0.3018/0.3731/0.0556/0.332 | 0.2203/0.2683/0.0381/0.2386 |
| public 4000, random walls | plain | 0.1443/0.1730/0.0226/0.183 | 0.1199/0.1441/0.0189/0.1435 |
| public 4000, random walls | prior | 0.0932/0.1085/0.0119/0.132 | 0.0657/0.0769/0.0077/0.0891 |
| our own training split | prior | 0.1250/0.1321/0.0170/0.171 | 0.0791/0.0811/0.0103/0.0947 |
| public 5000, classical cavity | plain | 0.2919/0.3711/0.0490/0.319 | 0.2444/0.3612/0.0484/0.3217 |
| public 5000, classical cavity | prior | 0.3092/0.4730/0.0561/0.328 | 0.2281/0.3498/0.0422/0.2584 |

Three readings, in order of how much they matter for practice.

- Within the family, the prior is what makes a single-$\varepsilon$ data set reusable. The identical architecture, budget and data give 0.1443 ($u_x$) with the plain output and 0.0932 with the prior—a factor 1.55—and 3.2× better than doing nothing, on $\varepsilon$ values away from the training contrast. It is even better than the same prior trained on our own 382 samples

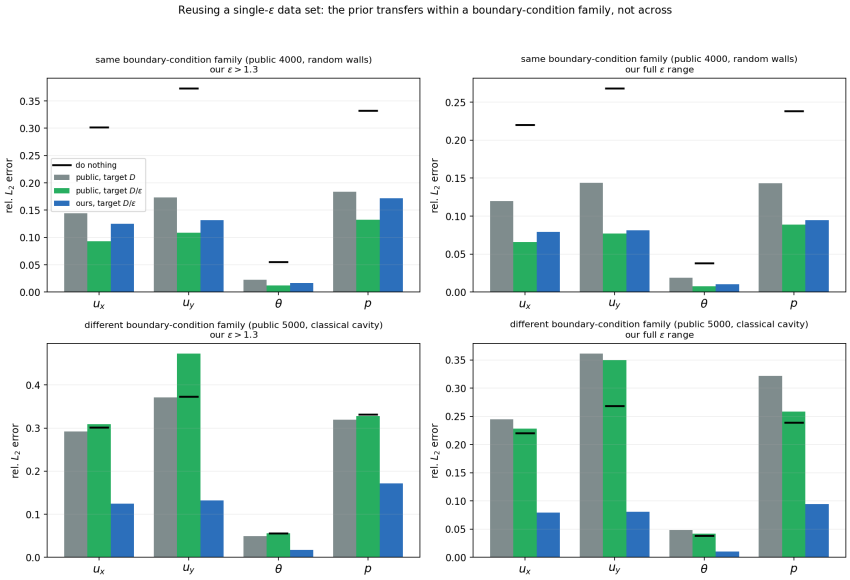



Figure 6: Reusing a single-$\varepsilon$ data set. Within the boundary-condition family the prior transfers and beats the plain model; across families it fails, and can be worse than doing nothing.

(0.1250), because it sees ten times more configurations. This is the concrete content of "the $\varepsilon$ dependence need not be in the data".

- The prior also fixes the failure mode of the plain model. The plain model's error at unseen $\varepsilon$ is largest where the training $\varepsilon$ is furthest away (its correction is a fixed field, Sec. 5.3), whereas the prior's error stays proportional to how wrong $P$ is, uniformly in $\varepsilon$.

- Across boundary-condition families the prior does not rescue transfer. Trained on the classical cavity, the plain model lands at 0.2919 against a floor of 0.3018—it learned little that transfers—and in extrapolation the prior is worse than the plain model in all four channels (0.3092/0.4730/0.0561/0.3280 against 0.2919/0.3711/0.0490/0.3192). In interpolation the prior does beat the plain model in all four channels (0.2281/0.3498/0.0422/0.2584 against 0.2444/0.3612/0.0484/0.3217), but it still does not beat doing nothing. The failure is the one Sec. 5.7 predicts: $A$ is a functional of the flow, and a surrogate trained on one family does not know another family's $A$. The classical cavity's peak Boussinesq speed is 41% above the random-wall median, so $\hat{u}$ separates the two families; it does not, however, order the per-sample error within the public random-wall splits under the per-channel relative $L_2$ metric used here (Spearman $-0.20$; partial correlation 0.09 at fixed $Ra$). We therefore use $\hat{u}$ as a family-level diagnostic, not as a

within-family transfer law.

### 5.10. A second geometry (aspect ratio 2:1, 104 samples)

Same generator, same ranges, cavity twice as wide (Fig. 7). The exponent is consistent with the unit-square fit in all four channels ($m$ = 1.09/1.16/0.93/1.08, within 0.3–1.8 combined standard errors); the two-parameter fit that adds a Rayleigh term is in Table S2, while the prefactor moves ($A$ = 0.175/0.175/0.008/0.222 against 0.147/0.193/0.010/0.148) and the Rayleigh trend strengthens for $u_x$ and $p$. The prefactor therefore has to be learned, which is why the surrogate takes $X$ as input rather than a table of $A$. Changing the fluid instead ($Pr$ 0.71 → 4.38) leaves the paired exponent within 1.5% and the paired median $R$ within 5%, with prefactors moving by at most 6.1% (Sec. 5.13).

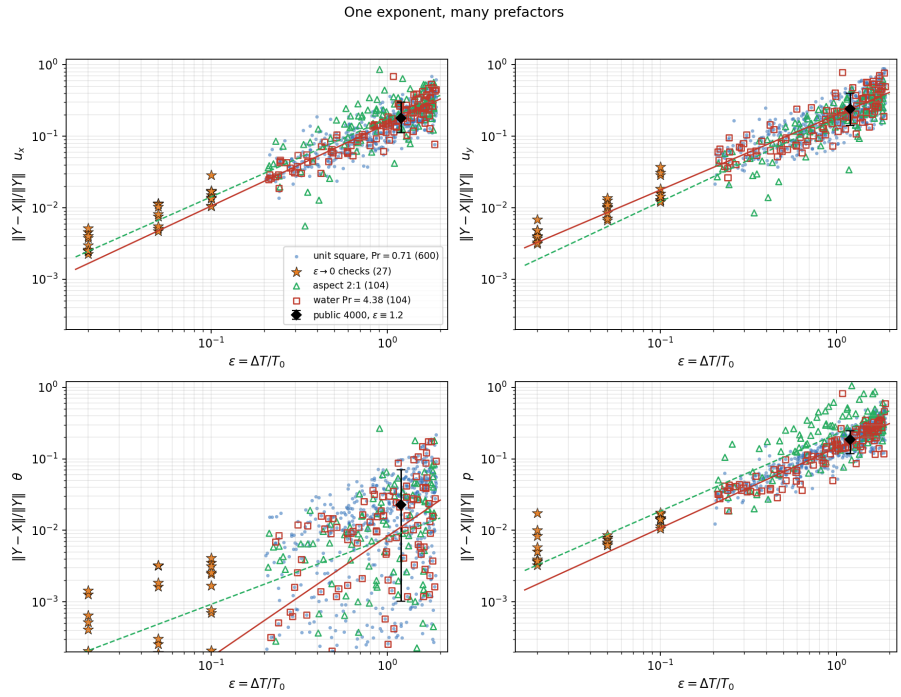



Figure 7: One exponent, many prefactors: unit square ($Pr$ = 0.71), aspect 2:1, $Pr$ = 4.38, and the public data set at $\varepsilon$ = 1.2.

### 5.11. Solver noise is not the source of the law

Over the 600 compressible members the median relative convergence change is $3.75 \times 10^{-6}$ (p90 $1.09 \times 10^{-4}$, worst $7.04 \times 10^{-4}$). Dropping the 65 samples above $10^{-4}$ moves the $u_x$ slope from 0.1809 to 0.1813, and the ten worst samples rerun at steady_tol= $10^{-8}$ are bit-identical: the march stops

on a zero-iteration guard, so the reported residual is the fixed-point residual of the discrete solution. Supplementary Sec. S10 collects the audit.

### 5.12. Cost and speed-up

A paired solve splits into a Boussinesq member (median 11.7 s) and a compressible member (22.3 s); the surrogate replaces only the second, so one pair falls from 33.9 s to 11.7 s, a factor 2.9. Because $X$ does not depend on $\varepsilon$, the amortised gain grows with the number of contrasts: one flow at 100 contrasts costs 16.6 s against $3.77 \times 10^3$ s (230×), and the 20-flow × 5-contrast scan measured here costs 321 s against $4.07 \times 10^3$ s (13×). The scan as actually run, which repeats the Boussinesq solve at every contrast, gives 17×; we quote the conservative same-reuse 13×. The solver runs on eight CPU threads and the surrogate on an RTX 4060 Laptop GPU (batch 1, 6.3 ms), so the two are not the same hardware. Table S10 collects the timings.

### 5.13. A third Prandtl number: $Pr = 4.38$

$Pr$ enters the coefficients of both members and hence $X$, but never multiplies $\varepsilon$ in the expansion. The prediction is asymmetric: the exponent must not move, the prefactor may. We generated 104 strictly paired configurations at $Pr = 4.38$ (same seed stream, $Ra$ and $\varepsilon$ as the first 104 air cases). The paired exponents agree within 1.5% (1.150 vs 1.157, 1.049 vs 1.047, 1.686 vs 1.668, 1.124 vs 1.109 for $u_x/u_y/\theta/p$), and the median ratio of $R$ is 0.95–1.01. The fitted prefactor moves by 3.3% ($u_x$), 0.2% ($u_y$), 1.2% ($\theta$) and 6.1% ($p$), far less than the geometry or boundary-family changes (Secs. 5.10, 5.7). Table S9 and Fig. S1 give the fits and the paired comparison. $Pr$ is therefore not a break-symmetry axis.

### 5.14. A three-dimensional check

Twenty-four strictly paired three-dimensional solutions ($res = 16$, $Pr = 0.71$; three wall types at $Ra = 10^4$ over $\varepsilon = 0.2$–$1.6$ and at $Ra = 3{\times}10^4$ over $\varepsilon = 0.4$–$1.9$) return window-local exponents for $u_x$, $u_y$ and $p$ of 1.03–1.14, within 0.15 of unity ($R^2 \geq 0.96$). The two Rayleigh numbers cannot be separated (0.065 against a combined standard error of 0.067), the prefactor varies by up to 13× between flows, and $\theta$ is reported separately. With one resolution and two Rayleigh numbers this is a consistency check, not a converged three-dimensional validation; the fits, the figure and the comparison with two-dimensional bins of the same window are in Supplementary Sec. S9.

## 6. Limitations

1. Two dimensions for the main sweep. The law is calibrated on 2-D solutions; a 2× mesh refinement moves the exponent by less than 0.005

at $Ra = 10^4$ and by at most 0.015 at $Ra = 10^6$, where the largest pointwise change in $R$ is 3.3% (Sec. 5.1). The 24-sample 3-D check of Sec. 5.14 returns exponents for $u_x$, $u_y$ and $p$ within 0.15 of unity at $Ra = 10^4$ and $3 \times 10^4$, but it uses a single resolution ($res = 16$); the POD-ROM baseline is a linear model with at most four modes, not a deep operator. Each 3-D pair costs 12–32 min of outer wall-clock time on this machine. A 3-D sweep at higher $Ra$ or finer resolution remains out of reach here.

2. $Ma$ is inert here, so one parameter is free rather than two. At larger $Ma$ this would have to be redone.
3. $\varepsilon < 2$, imposed by the equation of state (the cold wall must stay positive).
4. The power-law fits of Sec. 5.1 leave 16/18/78% of the variance unexplained in $u_x/u_y/\theta$ and 13% in $p$; the $\chi$ form of Sec. 5.1 raises $R^2$ from 0.19 to 0.78 and the $\hat{u}$ form of Sec. 5.7 to 0.90, and the same residual flow-dependence remains in the other three channels.
5. $\theta$ is not part of the common-exponent claim. Its free power-law fit is not identifiable at this sample size: $R^2 = 0.22$, slope 1.024 over the full window and 1.67 on the 104-configuration subset, and the fitted amplitude is not predictive (0.012 at $\varepsilon = 1.2$ against a public median of 0.0228). We make no exponent claim for $\theta$.
6. The pressure channel is quoted in the fixed gauge and hydrostatic correction of Sec. 5.2. Its exponent is not gauge-invariant, and a different post-processing gauge would change it.
7. The prefactor $A$ is configuration-dependent and must be learned: it moves with the geometry (Sec. 5.10) and by 1.66× ($u_x$) / 1.38× ($u_y$) / 4.05× ($\theta$) / 1.07× ($p$) between boundary-condition families of an independent data set (Sec. 5.7), while it is insensitive to the fluid ($Pr$, Sec. 5.13).
8. The field norm does not certify the wall-normal derivative. A prior trained on the field alone leaves a 2.6% integrated wall-$Nu$ error at $\varepsilon = 1.2$, worse than the 0.7% of doing nothing; the wall-flux penalty of Sec. 5.8 removes it (0.11%), but the loss has to be stated.
9. The earlier public release used a pinned $Fr = 1.095$, which puts its $Ra_{\mathrm{eff}} = Ra\,Fr^2/\varepsilon$ 0.08% below ours at $\varepsilon = 1.2$; the v2 release corrects this.

## 7. Conclusion

The object a compressible-convection surrogate has to reproduce—the gap between the Boussinesq and compressible solutions—factorises: $\varepsilon$ sets its leading-order scaling, and the flow sets its amplitude and spatial shape.

1. One break-symmetry parameter. In this non-dimensionalisation $Fr^2 = \Delta T/T_0$ identically, so the axis is $\varepsilon = \Delta T/T_0$; Mach-number effects enter at $O(Ma^2)$ in the retained equations and are below the resolution of the sampled range. The amplitude is not universal: $P(X)$ carries the flow dependence.
2. A leading-order law. $R = A\varepsilon^m$ with $m = 1.023 \pm 0.018$ and $0.995 \pm 0.019$ for $(u_x, u_y)$ and an intercept consistent with zero over the full two-decade window; from $\varepsilon = 0.8$ upwards the same points give 1.23–1.32, the $\varepsilon^2$ curvature. A 134-family boundary-condition description adds only 2.3–3.3 pp of adjusted $R^2$ to the velocity and pressure channels. The fit leaves 16/18/78/13% of the variance unexplained in $u_x/u_y/\theta/p$; $\theta$ collapses on the advection strength $\chi$ rather than on $\varepsilon$ alone and carries no exponent claim. Pressure gives $m = 1.009 \pm 0.015$ in the fixed gauge. Twenty-four paired 3-D solutions at two Rayleigh numbers are consistent with the same near-unity exponent for $u_x$, $u_y$ and $p$ (within 0.15 of unity).
3. A reason and a design rule. The $\varepsilon$-expansion reproduces the Boussinesq system at $O(1)$ and produces the correction through $\nabla \cdot \boldsymbol{u}_1 = \nabla \cdot (\theta_0 \boldsymbol{u}_0)$ and the density-modulated inertia; the linear operator contains no $\varepsilon$, so $D = \varepsilon P(X) + \varepsilon^2 Q(X) + \ldots$ with a shape that does not rotate. One Newton step of the retained residual at $X$ reproduces the measured correction to 4.7/3.8/7.2% (median, $u_x/u_y/p$) on the $\varepsilon = 0.02$–$0.10$ sweep. Predicting $P(X)$ and multiplying by $\varepsilon$ outside the network has 1.09× lower channel-averaged error than a FiLM-conditioned network in extrapolation ($\theta$ and $p$ individually, $p$ in the fixed gauge; the $u_x/u_y$ intervals include zero), transfers to a Fourier neural operator, and makes a public single-$\varepsilon$ data set reusable: 0.093 ($u_x$) on $\varepsilon > 1.3$ against 0.144 for the plain model and 0.302 for doing nothing. That reuse holds within a boundary-condition family and fails across one, predictably. The end-to-end saving is 2.9× per pair and 13× for the 20-flow × 5-contrast scan.

The heat-transfer audit adds a constraint that the field norm hides. At $\varepsilon = 1.2$ the public prior lowers the temperature error from 7.2% to 0.10% and the local wall-flux error from 6.0% to 4.3%, yet its integrated wall-$Nu$ error is 2.6% against 0.7% for doing nothing. A relative $L_1$ penalty on the wall-normal gradient brings the integrated error to 0.11% and the local error to 1.0%, so the failure belongs to the training target rather than to the factorisation. For the isothermal-sidewall family the integral correction is configuration-dependent and changes sign between $\varepsilon = 1.2$ (+0.51%) and 1.9 (−1.89%).

Section 6 lists the limits: two dimensions for the main sweep, $\varepsilon < 2$, and a prefactor whose power-law fit leaves 16% ($u_x$), 18% ($u_y$) and 78% ($\theta$) of the variance unexplained; the $\theta$ gap closes to 22% once $\chi$ is supplied. The

leading $\varepsilon$ factor is derived; the amplitude, the window-averaged exponent and the $\varepsilon^2$ curvature are measured.

## 8. Reproducibility

Table S7 maps each script to the results it produces. The solver, surrogate and reporting scripts (dolfinx 0.7.3, PyTorch 2.10) and the machine-readable results behind every table and figure are mirrored in the repository listed under Data availability. The raw solver output (~6 GB: 600 random-wall, 104 aspect-2:1, 104 $Pr = 4.38$, 27 $\varepsilon \to 0$, 100 paired and 10 tight-rerun samples, ten $128^2$ mesh pairs and 24 three-dimensional pairs) stays on the machine described above. No number is transcribed by hand.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used an AI assistant (OpenAI Codex) to develop analysis scripts, to draft and edit the manuscript, and to check numerical consistency between the text and the stored results. After using this tool, the authors reviewed and edited the content and take full responsibility for the content of the publication.

## Data availability

The public paired data set of Menglik et al. [5] is available under CC BY 4.0. Our generated data, solver configuration, and every script that reproduces the numbers in this paper are available at https://github.com/sheep-cloud12138/Algorithms-instead-of-simulations (branch dev); the raw solver output is available from the authors on request.

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